

Topic: Queue using Singly Linked List

Q. Describe the implementation of a Queue using a Singly Linked List. Detail the logic for the Enqueue and Dequeue operations.

****Definition to Remember****

A Queue
avoids t
**`rear`*

Queue Implementation using Singly Linked List

To achieve $O(1)$ time complexity for both Enqueue and Dequeue, we must maintain references to both ends of the list.

| Queue Pointer | Linked List Equivalent | Used For |

| :--- | :-

2. Enqueue Operation (Insert at End)

Adds a new element to the rear of the queue.

Algorithm:

text

3. Dequeue Operation (Delete from Beginning)

Removes the element from the front of the queue.

Algorithm:

text

4. Advantages of Linked List Implementation

* ****Dynamic Sizing:**** Queue can grow infinitely (bounded only by heap memory). No "Queue Full" fixed limit.

* ****O(**
instant.

*****Must-Write Points (for 10 marks)*****

1. Queue

*****Quick Recall*****

`Queue`
rear = n